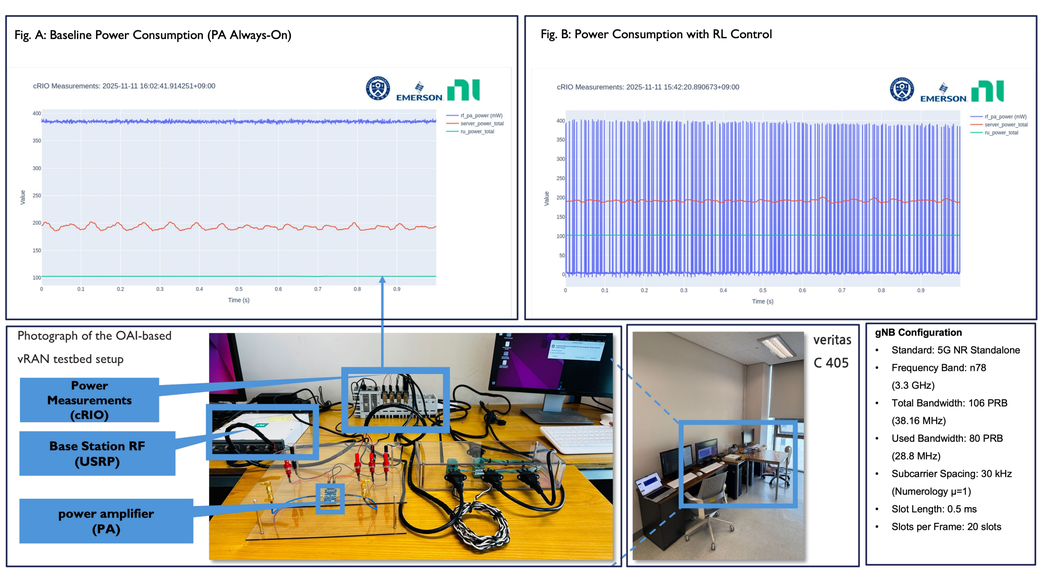
Real-time 6G Wireless Systems

@IEEE GLOBECOM 2025, Taipei, Taiwan

Intelligence Networking Lab., School of Integrated Technology, Yonsei University, Korea
https://www.cbchae.org | cbchae@yonsei.ac.kr

System 1: Reinforcement Learining-Based Energy-Efficient RAN: Algorithms and Prototype Validation

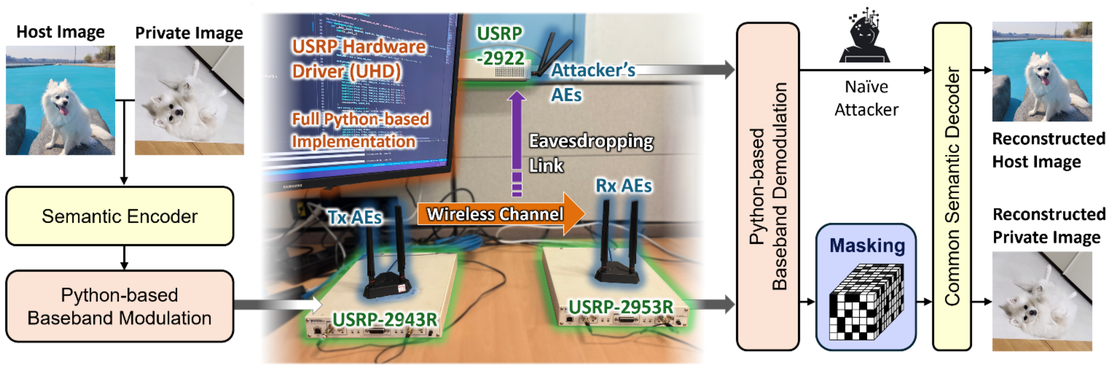
Hajin Park, Sangha Park, <NI members>, and Chan-Byoung Chae

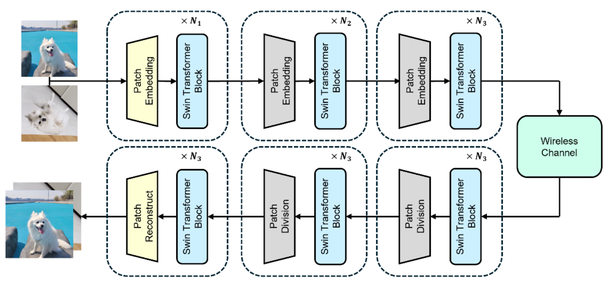


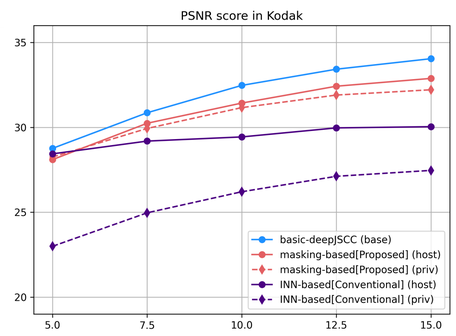
- Real-time PA power optimization for 5G vRAN using Deep Reinforcement Learning (DRL)
- Hardware-in-the-loop 5G SA stack (OAI) with real-time state acquisition and slot-level PA control
- Live testbed validation integrating NI hardware (USRP X410, cRIO) for synchronized QoS and power measurements.

System 2: Secure Semantic Communications with Steganography

Dongha Choi, Hanju Yoo, and Chan-Byoung Chae



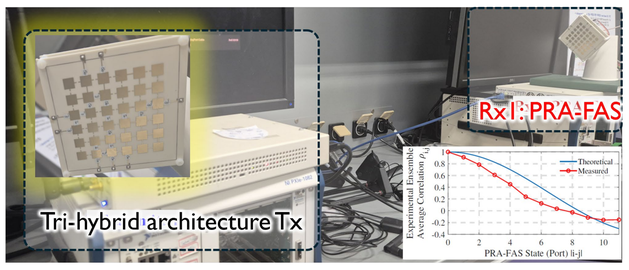


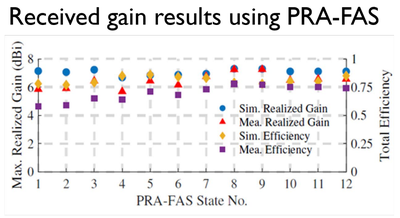


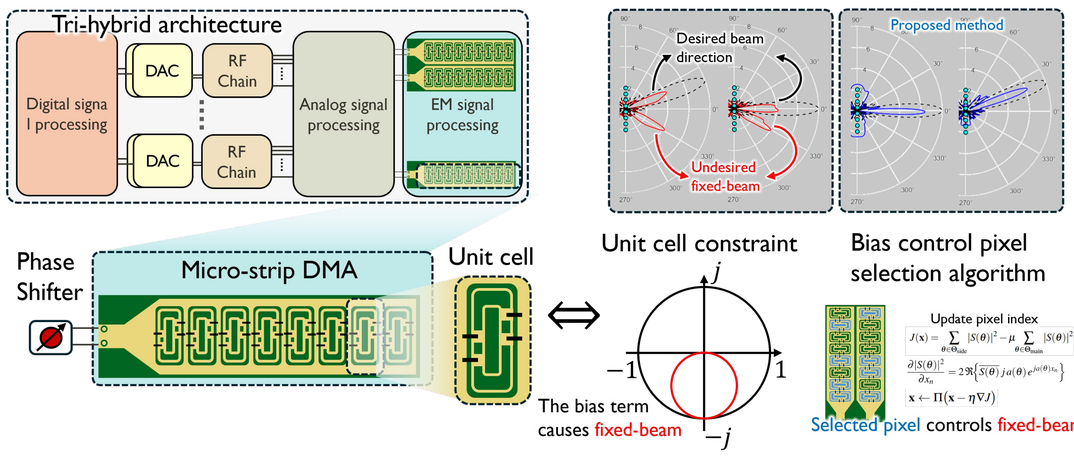
- Real-time demo of image (video) semantic communications
- Secured transmission system with masking-based steganography module
- Neural network architecture that facilitates adaptive rate control and flexible mask reconfiguration
- Fully python OFDM-MIMO stack with ZF-based semantic symbol multiplexing

System 3: Fluid Antenna System with Tri-hybrid MIMO

Yoontae Kim, Siyun Yang, Haewoong Hyun, Kai-Kit Wong, K.-F. Tong, R. Murch, and Chan-Byoung Chae







- Solve the limitations of conventional beamforming and massive MIMO
- Real-time 6G MIMO demo with novel Pixel-based Reconfigurable Fluid Antenna System μs-level fast switching across 6 states matching ULA correlation
- Proposing energy efficient and high-performance tri-hybrid architecture

* Pixel-prototype by T. Liang and Prof. Tsang from HKUST GZ
* 6.6 dBi gain and ~80% average efficiency in all states

- This work was supported by the Ministry of Science and ICT and the IITP (**RS-2024-00428780**, RS-2022-0-00704, RS-2024-00404972 , RS-2024-00453301, ERC2022-0-00704).
- Full demo videos are available at https://www.youtube.com/@chaeyonsei

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System 4: <SensorView Products>

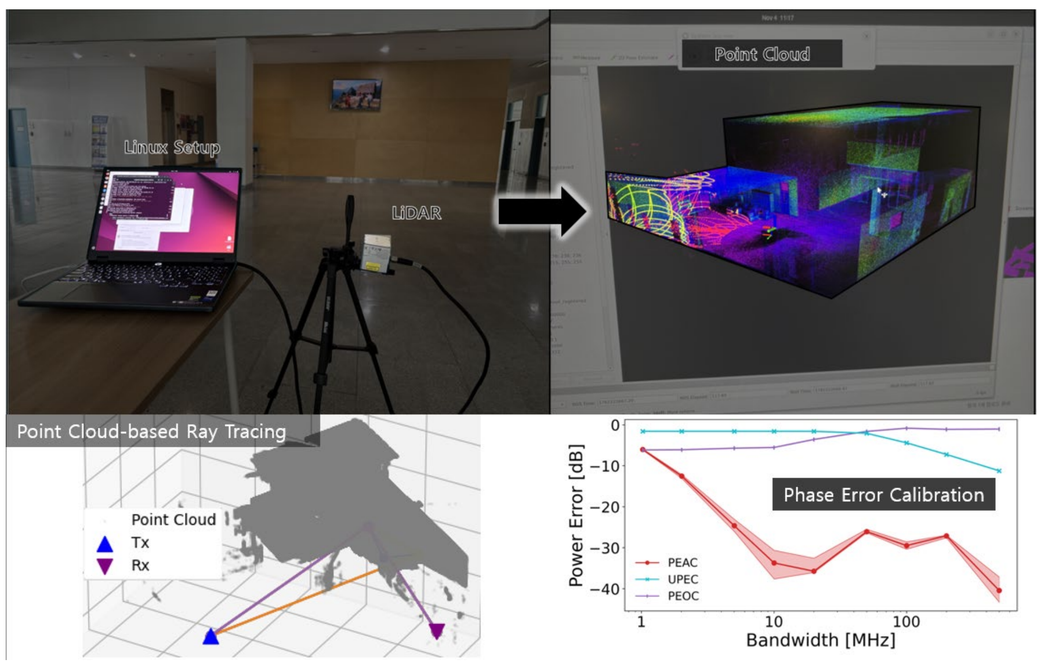
Byoungnam Kim, and Chan-Byoung Chae



- Place holder
- 센서뷰에 연락해서 관련 내용 달라고 요청드렸습니다.

System 5: Point Cloud-based Power Error Calibrated Hyper Realistic Digital Twin

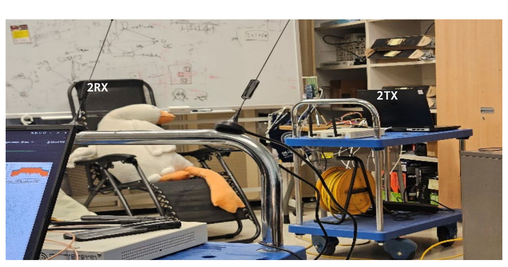
Hoyeon Park, Seokki Min, and Chan-Byoung Chae



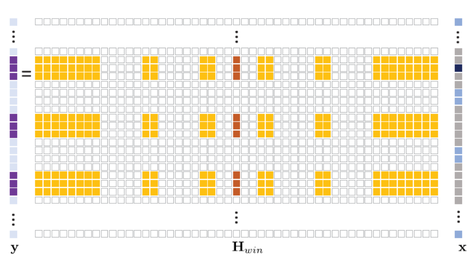
- LiDAR-based point clouds are utilized to generate high-accuracy 3D maps
- A phase error calibration method enhances the precision of the channel state
- Hyper-realistic ray tracing simulations can be conducted by integrating these two preceding component

System 6: Practical Implementations and advances of Next-Generation waveforms

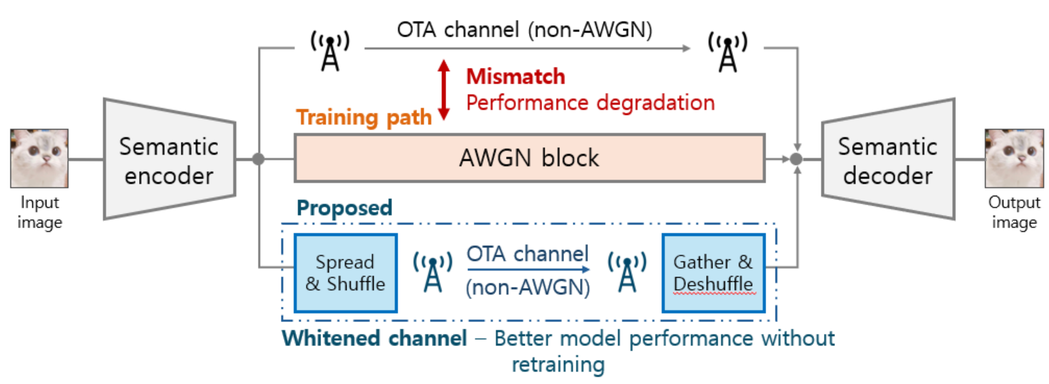
Minjae Ko, Hanju Yoo, and Chan-Byoung Chae



MIMO-OTFS Testbed Setting



MIMO-OTFS Pilot Embedding



- Robust OTFS-MIMO implementation with doubly-selective environment assumptions
- OTFS-style precoding method for Gaussian channel estimation and symbol correlation reduction
- Improved PSNR by shuffling subcarriers
- Semantic-Communications adapted on OTFS-MIMO

- This work was supported by the Ministry of Science and ICT and the IITP (**RS-2024-00428780**, RS-2022-0-00704, RS-2024-00404972 , RS-2024-00453301, ERC2022-0-00704).
- Full demo videos are available at https://www.youtube.com/@chaeyonsei